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CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Project Interim Report
[Safe Shift]

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1.0 Introduction

Information Technology is an industry that specializes in innovation as well as productivity, allowing development of new technologies and solutions to solve real-world problems. IT has rapidly transformed nearly every aspect of our lives from communication to entertainment. Complete I.T. (2020). Emerging trends such as AI and quantum computing are set to redefine the industry driving innovation and connectivity. Globaltechaward.com. (2025).

My project is called safe shift it is a prototype consisting of core purposes of providing accurate forecasts and alerting warnings as well as efficient data to users as a mobile app. The prototype focuses on the innovation aspect of IT being a new technology/solution to solve ongoing problems. Most weather apps are criticised for inaccurate data and late warnings for severe weather which damages peoples plans and puts them at risk

1.1 Problem Statement

Data inaccuracy Is a common problem most weather apps face today, many of them are criticised by the public for inaccurate or insufficient data making this an ongoing problem. Many severe weather warnings go out late to the public affecting user plans and increasing risk. Climatecosmos.com. (2025). My project aims to solve the problem: false sense of security problems through weather alerts. When severe weather strikes without warning people aren't prepared to take shelter or evacuate safely which is a common risk. My project aims to give alerts to people when severe weather strikes so users can evacuate safely whilst also having optional features such as schedule/calendar integrations allowing users to integrate plans meaning when alerts do happen the product can cancel the plans and reschedule it aligning with the principle of safety.

2.0 Review of Existing Knowledge

Based on my research and papers I have identified, existing technologies are out there close to what my prototype plans to be. The sources have been critical to finding existing technologies and knowledge and talk about current and emerging technologies.

Machine Learning

Machine Learning methods are quite common in terms of using algorithms to predict weather forecasts. [Sayor et al., 2025] uses an application to provide users with real-time weather forecasts utilizing IOT sensors and machine learning algorithms to collect comprehensive data

such as humidity, windspeed and barometric pressure. Climate change increases frequency of disasters like hurricanes meaning data tools such as IOT sensors and machine learning algorithms are crucial for disaster preparedness minimizing loss of life and property [Sayor et al., 2025]. Climate change is a major challenge this century as it's not easy to predict when the weather could change due to insufficient data. [Guardia et al., 2025] refers to how specialists used a 3D WebGIS to map precipitation using machine learning algorithms in the region of Palermo (Italy) being developed by JavaScript libraries and using python scripts for activating data connections. API integration, real-time data processing and IOT automation tools are used on technologies that deploy the same idea as mine with built in alerts. [Sandhip et al., 2025] refers to how OpenWeatherMap is used for dynamic mapping and real-time alerts based on location. Machine Learning algorithms have been used to predict crop diseases through real-time weather with different models being used to predict yellow rust algorithms [Biljwan et al., 2025]. ML models are used to generate air temperature using historical data fetched from OpenWeatherMap [Juneja et al., 2025]. Data fetched from these models have histories of weather events within different meteorological elements like humidity and atmospheric pressure [Juneja et al., 2025]. ML Models trained on datasets improved efficiency through metrics such as MSE and RMSE, the metrics presented quantify the difference between predicted temperatures and their actual values providing a clear picture of how accurate the model is fitted [Juneja et al., 2025]. Random Forest machine learning algorithms have been developed on Raspberry PI 3 which gets data from humidity and pressure sensors to predict rain [Singh et al., 2019]. Different modules are used in which the ML model collects data from [Singh et al., 2019] meaning there are varieties in which it can choose from effectively finding different data. ML models have also been used to map accurate seasonal precipitation forecasts for the Western United States using ML multi-models. [Pinheiro and Ouarda 2025]. ML Models like Pangu-Weather and GraphCast struggle to reproduce sub-synoptic weather phenomena leading to forecasts that become overly smooth over time and in some cases ML models lack accurate representation of balances in the atmosphere.

Application Programming Interface (API)

API technologies are perhaps the most common source in terms of gathering accurate weather forecasts. [Chadha et al., 2024] studies aim to create a robust app which provides real-time data to users, in which data is collected from the software OpenWeatherMap and queried using HTTP requests with data fetched in JSON. The data includes current weather conditions, hourly forecasts and daily forecasts with integration parsed by JSON responses stored in a database [Chadha et al., 2024]. OpenWeatherMap provides reliable weather data and historical information crucial for a weather app. Forecasts are made by collecting as much data as possible about the current state and understanding atmospheric processes, this forewarns people about future conditions so people can plan accordingly. Once applications are built you get Weather API which uses python source codes to get real-time weather data [Dewi and Chen, 2019]. OpenWeatherMap provides an API with JSON, XML and HTML endpoints as well as limited free

user tiers. Historical data access requires a subscription of $150 p/m in which users request current and extended forecasts with maps showing wind speed, pressure and precipitation [Dewi and Chen, 2019]. [Hermanth et al., 2024] refer to how apps leverage technologies like HTML, CSS, JavaScript and integrate OpenWeatherMap API to provide updates for user locations. API in particular processes data using Machine Learning to enhance numerical weather prediction models provided by sources such as NOAA [Musah et al., 2022]. Whilst APIs are common, they also produce inaccurate reports and are costly if you need historical information, not all APIs will provide accurate forecasts.

Agile Methodology

Agile Methodology is the most common approach used for problem solving and in this case software/application development. The approach originates from lean manufacturing in the 1940s [Spartu, 2010]. The comparisons of agile and mobile application features include documentation, planning, development team size, scale and customer identification [Spartu, 2010]. Agile helps companies build the right product and empowers teams to continuously redesign their release to optimize value through development allowing them to be competitive in the market [Flora and Chande, 2013]. It focuses more on human interaction over tools and processes considering core values such as interactions, working software, customer collaboration and responsive change [Flora and Chande, 2013]. Agile breaks down programming, development and project management into smaller sub problems compared to waterfall methodology where the requirements are defined at the start in which users find it hard to describe what they need the app to do [Patil et al., 2016]. Agile assesses each phase giving direction to the project to shape up a final product [Patil et al., 2016].

Alerting Systems

Integrated alert systems are becoming increasingly popular with Smart weather monitoring and alert systems using IOT sensors [Rahut et al., 2018] refers to an app that sends notifications as an effective alert system to warn people about sudden weather changes. Different modules are used on Arduinos and sensors connected by microcontrollers to measure different things. LDR sensor to measure light intensity and ML8511 module using ultraviolet light sensors to warn users of sunburn. [Rahut et al., 2018]. System nodes are installed into sensors for data collection and transmitted through a multi-hop network which is processed and sent to people in affected geographic regions via SMS [Shiravale et al., 2015]. Radio stations in the United States like NOAA weather radios provide alerts on the radio to users. [Collins and Kapucu, 2008] explains how siren systems are used across the Midwest region of United States alerting communities of impending weather disasters with a loud blare. NOAA weather radios broadcast continuous weather forecasts from nearby offices which activates whenever an alert is sent out however, not many people own a NOAA radio meaning they miss out on potential weather alerts for

severe conditions so if a transmitter loses power other disaster alerts may not be sent out in a timely manner [Collins and Kapucu, 2008]. IOT and Machine Learning technologies have been used to warn users of severe conditions using hardware sensors to measure aspects like air pressure, humidity and carbon monoxide in the atmosphere [Durrani et al., 2019]. These algorithms were prepared using NARXNET technology to check the predicted data to see if it fulfils the conditions of an alert, if that is the case it generates alert for disasters such as heat stokes and rainstorms [Durrani et al., 2019]

Location Technologies

Location-based services aim to provide information to users based on geographic area at any time [Dao et al., 2002]. Pinpointing specific location is challenging which is why non-GPS solutions use COO (Cell of Origin). COO is straightforward as it uses cell identification for approximate location of the caller [Dao et al., 2002]. LBS is a major requirement for a GPS locator as this allows users to get their own current location and view on google maps. Locations are calculated with built in GPS with coordinates on maps using third-party API called MIDMAPS which is developed through JAVA 2.0 and uses MySQL for database hosting [Chandra et al., 2011]

Existing Applications

Similar apps like AccuWeather are known to be some of the best out there but lack alerting systems. AccuWeather allows push notifications which can be turned on/off by users, these notifications are location specific instead they're radius specific sending alerts within 10 miles of a user's location. NOAA radio stations provide alerts for anyone who has them installed but not for users that don't have them increasing risk of safety for them. Disagreement amongst people is common when receiving multiple weather updates from different sources. In October 2016 historic and destructive winds were forecasted for 15th October with initial warnings being sent on Wednesday and by late Friday the chance of it happening was decreasing [Burgeno and Joslyn, 2020].

2.1 Critical Analysis and Gap Identification

Based on the sources identified existing technologies have been used to create similar applications involving machine learning and API integrations but there are a lot of faults, some of these faults include the inaccuracy of data forecasting. Whilst recent technologies and projects have shown good progression with these integrations Apps like AccuWeather have been heavily criticised for inaccuracy, as well as the push notifications not working as well as it should be. Many weather applications use notifications, but these can be turned on or off even

displaying radius data or insufficient data. The gap identification being using effective alert systems to alert users about potential severe warning in their geographical area instead of notifications 10 miles away.

While Machine Learning algorithms are great the downsides to them are that they require high quality and extensive datasets also having critiques in ML models being expensive requiring significant investment in infrastructure. API integration has its limitation which include rate limiting. Exceeding these limits result in additional charges or denial of service. Even with all the negatives of existing methods the clear gap I have identified is regarding in alerting systems while we can expect some sort of data inaccuracy the gap involving safety remains an issue. NOAA radio stations provide real-time forecast accuracy though broadcasts however, the downside to this technology is that not everyone in the US owns a NOAA radio station so getting severe weather alerts to them is seemingly a challenge. Agile Methodology is a commonly used approach where people break down larger parts into smaller sub parts to create a valuable outcome, but this comes with costs such as unstable planning and misinterpretation of feedback from clients.

2.2 Project Justification

The justification of my project is clear whilst there are many advantages in which applications use these technologies, they haven't solved a critical gap regarding safety which is what my project will do. Based on the critical analysis the gap is the use of alert systems for safety. This is a common issue, and similar technologies only utilize push notifications as an alert system. This makes my project relevant as the intended prototype will build on this problem using alert warning systems based on a user's specific location instead of nearby locations. This is what many existing apps lack as they use location technology for severe alerts based on the user's nearby city for example 10 miles. Push notifications can be turned on/off meaning users can miss severe weather forecasts, my app won't allow that as it will directly alert the user with a beep like how they use systems in government tests for threats etc. This is a gap that has not yet been explored and is a core purpose to why it is relevant, it improves safety and allows users to plan accordingly based on alerts with scheduling and cancellation systems rescheduling activities based on the alerts. Aside from using the location and alerting system technologies the main development will consist of existing technologies such as machine learning and API integration however, my project aims to use these to pull accurate efficient data instead of inaccuracies some applications have. As a summary to why my project is relevant, many apps have forecast inaccuracies and fail to implement warning alert systems to alert people of severe weather causing tragedies such as loss of life.

3.0 Project Aims, Objectives, and Scope

3.1 Project Aim

The Aim of my project is to develop an application to solve the false sense of security problems through weather apps, by using integration tools such as API keys to pull real-time weather data and location-specific technologies to create customized weather alerts based on users coordinates whilst also having automatic cancellation systems built in to reschedule or cancel a user's plans based on severe weather conditions.

3.2 Project Objectives

- *Research into Existing Technologies / Problem Identification*
- *Evaluate Tools and Software for Prototype Development*
- *UI/UX Designing / Model Development*
- *Designing and Developing the Back End of Application*
- *Deploy and Implement the Application / Evaluation*

3.3 Project Scope

This Project will focus on Software Development and Integration methods. It will include numerous software applications to help build the front-end design UX part of the application, as well as Integration protocols with AI and API keys to help pull the real-time data. This project won't include any hardware materials to pull data weather as it is strictly application and software based. The project will include many features when completed including data management, testing and user interaction however, it won't include advanced or complex data formats as this makes it unrealistic for me and the project. Overall, the project concentrates on user interaction using Innovative Protocols and Solutions and will not focus on any complex integration.

4.0 Project Design

4.1 Design and Methods

Objective 1

The design & methods for this prototype will link to the project objectives and how I will achieve each one. Objective 1 will focus primarily on data collection methods, methods such as qualitative data like interviews to see what problems people face with current applications. Secondary research is to also be conducted as this will focus on finding existing technologies used to create similar applications.

Objective 2

Once the research is conducted from objective 1 the next step is evaluating tools and software's for the prototype design. This objective will include research such as finding the software's I'm going to use for the development. Once the qualitative data from objective 1 is completed this will narrow down the solution design helping me research to find the appropriate tools to develop the prototype. The way I plan to complete this objective before researching software's is conducting interviews in objective 1 where users can explain what features they want in the app, how they want it to look, colour scheme etc. This data will provide insight into finding the appropriate development tools.

Objective 3

Once the research and tools are conducted the designing begins. The prototype will be created based on the data I have received creating something like what the people want. This objective can only be completed if objectives 1 and 2 go to plan. This phase will also involve existing technologies that will be built into the prototype to make it better.

Objective 4

Directly links to objective 3 which is just the development stage of the prototype based on research conducted and collected through objectives 1 and 2

Objective 5

The evaluation stage, this objective will consist of me using quantitative data collection methods for people so they can review my prototype and rate features such as functionality, features, text size etc. This method will most likely be a survey but could also be a questionnaire. Comparative

studies allow me to compare my app to other existing ones effectively seeing similarities and differences.

4.2 Justification of Methods

Objective 1

Qualitative data collection methods will be used as it allows me to gather specific knowledge from user's based on the problems they face and what they want on the prototype interviews are perfect for this stage because the responses I get will be in detail allowing me to effectively narrow down a problem and solution. Also researching into other technologies for this stage allows me to find methods and similar solutions I can use to develop my prototype having resources to fall back on. Secondary and Qualitative data research is the best approach for this stage as you can gather in depth responses and find existing technologies for solutions.

Objective 2

The reason I have chosen to research tools and software's is so that I can find the appropriate ones used for development, research is the only way to go about this objective as I plan to spend time using and testing out a variety of software's to see the best fit. The software chosen will link back to the interviews in objective 1 in which I find software's that are suited to the user's needs for example which software has the best design, best features etc. In terms of this objective there is no other way to go other than conducting your own research for development.

Objective 3

The reason no data collection methods or research will be used here is because they have all been conducted through objectives 1 and 2. This objective is the development where I use the resources and data, I have gathered to develop the prototype effectively using the qualitative data collected to develop the prototype to how the users want it to look. The use of researching into existing technologies plays a big part in this objective as I will plan to use these technologies in my prototype.

Objective 4

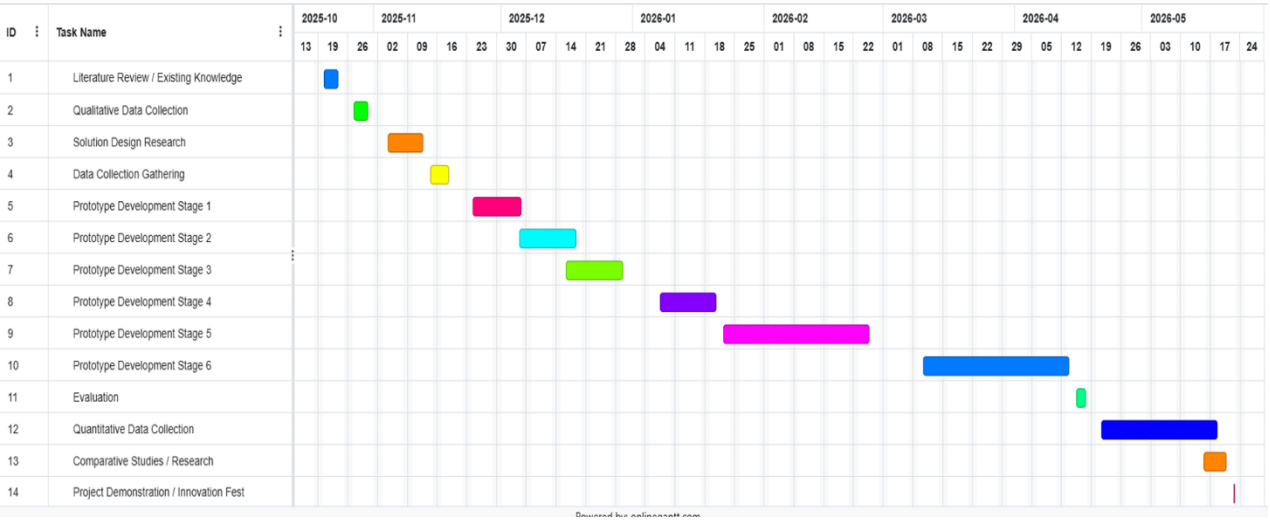
Carries on witlookse development from objective 3, no data and research carried out during this stage as the data has already been collected, now its just a case of me continuing to develop the prototype which solves the issues and problems users specified from the interviews and making

the prototype look to how they want it to be also again from the interview data. The research of existing technologies from previous objectives allows me to integrate similar methods into the prototype meaning the methods and research conducted earlier into the plan will be effective as the prototype can be created with resources and data using a strategic approach to development based on gathering.

Objective 5

Uses the most data collection methods, as this is the final stage/objective data collection methods such as quantitative data will be used, these methods are numerical making users answer specific questions about the prototype on how they found it. Surveys and questionnaires are the best approach for this objective as this will give users questions in which they pick a number for each one for example a question about the functionality with 5 being the highest and 1 being the lowest etc. Questionnaires use the same approach as the surveys allowing me to gather responses and see what parts of the prototype users liked and what can be improved. Comparative studies is also an effective approach I plan to use as this will compare my finished prototype to other existing applications out there in which I can see the similarities and differences between them, taking into account what I can do better next time whilst also seeing what I did better than certain apps and what I done worse.

4.3 Project Timeline



5.0 Feasibility, Risks, and Ethical Aspects

5.1 Feasibility

My project is realistic and achievable across 24 weeks as it's a prototype not a fully functional application. I believe my prototype is achievable because of the number of resources I have available allowing me to research specific technologies and existing apps to develop the prototype, resources such as conference papers, software tools, existing technologies and data collection methods account to this, the time being 24 weeks also gives this a realistic aim of it being achieved as a lot can get done in this frame such as research, data collection and development. I am confident that this can happen due to my knowledge and experience in research as I have all the things needed available like necessary data and software tools as well as being able to gather more resources based on available time.

5.2 Risk Analysis and Mitigation

Potential Risk	**Likelihood (Low/Med/High)**	**Impact (Low/Med/High)**	**Mitigation Strategy**
Lack of Feedback for Prototype Design/Development	*Medium*	*High*	*Gather enough feedback from potential users and stakeholders*
Project Planning/Project Management Risks	*Low*	*High*	*Set realistic timelines and objectives to achieve*
Poor Platform Selections for Designing	*Medium*	*High*	*Research and test out different software's to find the best solution*
Scope Creep	*Low*	*High*	*Create project management framework to manage goals and rigorous changes to development*

Table 4.1: Risk Assessment and Mitigation

5.3 Ethical Considerations

As I'm working with human participants, I will need secured consent forms for data collection so I can use their responses. This will be done through consent forms in which they read information then agree to sign meaning I have their consent to use their data. Ensuring anonymity for participants will be done by not collecting data such as names, phone numbers etc, that data is for them. Instead, each participant will have an ID Number, whilst also clarifying with the participants that their responses will be anonymous Ryerson University (2015). The government data protection act 2018 control how personal information is used by organizations.

As I'm using the data I must follow the guidelines like using the information fairly, don't share the data with anyone and don't keep data longer than necessary Legislation.gov.uk. (2018).

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